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Getting 'What Works' working: building blocks for the integration of experimental and improvement science

Amelia Peterson

Graduate School of Education, Harvard University, Cambridge, MA, USA

ABSTRACT

As a systemic approach to improving educational practice through research, 'What Works' has come under repeated challenge from alternative approaches, most recently that of improvement science. While 'What Works' remains a dominant paradigm for centralized knowledge-building efforts, there is need to understand why this alternative has gained support, and what it can contribute. I set out how the core elements of experimental and improvement science can be combined into a strategy to raise educational achievement with the support of evidence from randomized experiments. Central to this combined effort is a focus on identifying and testing mechanisms for improving teaching and learning, as applications of principles from the learning sciences. This article builds on current efforts to strengthen approaches to evidence-based practice and policy in a range of international contexts. It provides a foundation for those who aim to avoid another paradigm war and to accelerate international discussions on the design of systemic education research infrastructure and funding.

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Introduction

'What Works' is a colloquial term for an experimental approach to applied research, centring on the conduct and amalgamation of results from randomized controlled trials (RCTs) of educational interventions.[1] It is currently the most high-profile form of investment in education research in England, and finds increasing favour in Australia (Brown 2013; Bush 2014). In the U.S., however, where the approach has had longer to mature, major evaluation bodies are currently re-addressing their own practices, and alternative approaches to systemic research funding in education are increasing in profile and support. These alternatives, most prominently improvement science (Bryk et al. 2015; Langley et al. 2009; Lewis 2015), seek to address some of the limitations of 'What Works', in particular questions around the sustainability and scalability of practice. Like 'What Works', these approaches demand a high degree of funding support and time and energy from schools. For a centralized research infrastructure, therefore, there is a need to consider how to balance apparently competing approaches.

In this paper, I describe how the core elements of experimental and improvement science might be combined to complement each other's weaknesses. The key weakness of 'What Works' is that it generates evidence relevant only to single either-or decisions (Cartwright and Hardie 2012). The interdependence of factors in school contexts mean that actors are rarely in a position to make those decisions. In contrast, improvement science and other networked inquiry approaches are designed to fit with the daily context of teaching and learning, allowing for continuous small

CONTACT Amelia Peterson ameliapeterson@g.harvard.edu Graduate School of Education, Harvard University, Cambridge, MA, USA

decisions based on feedback and adaptation. These decisions need to be informed by reliable general principles. To supply these principles, a reframed 'What Works' strategy would focus on identifying a set of 'building blocks' in the form of key mechanisms of educational change. I describe how research initiatives in both the experimental and improvement paradigm could join in a common endeavour to identify, refine and test these building blocks, thus focusing attention on the most proximate factors that promote learning, as opposed to what is amenable to programmatization and trials. I hypothesize that this focus would improve the efficiency and applicability of experimental research in education.

This endeavour is proposed with caution as, alongside improvement science, there is no paucity of proposals for strengthening the utility and usability of education research (e.g. Brown 2015; Granger 2011). These proposals, however, tend to focus on increased attention to context and refinement, and rely on expert use of research at the practitioner level. This takes time, resources and expertise that may not be available. This paper focuses instead what could change in the design of trials to make their output more useful and useable.

Approach

This paper is based on a review of literature on evidence creation, evidence-based practice and research partnerships in education. I conceptualize the topic broadly as efforts to make the work of professional researchers and evaluators influential for educational practitioners and decision-makers. In this paper, following Brown (2013), I see research, knowledge and evidence as terms the applied research field uses somewhat interchangeably, to refer to claims that intend to legitimately inform decision-making (even if that legitimacy might sometimes be contested). Informing decision-making is of course only one of the ways that research might impact the work of educators (Weiss 1979, 1988). It is, however, the notion of research, knowledge and evidence that seems to be built into 'What Works' efforts, and so is the most applicable for this paper.

I also draw on conversations with managers, evaluators and users of major 'What Works' initiatives on each side of the Atlantic, predominantly those engaged in the Investing in Innovation (I3) grants in the U.S., and the Education Endowment Foundation (EEF) in the U.K. This paper is written to further discussions within and across these initiatives and ones like them.

Background

'What Works' as a paradigm for improving education

'What Works' is a label used to describe an approach to applied research in various policy fields. In the context of Education, 'What Works' can be summarized as a set of assumptions about the roles of system actors, researchers and schools:

(1) *System actors* are responsible for providing funding for 'trials' of promising programmes or practices; for aggregating the results of trials; and communicating aggregated judgments of impact to a wider education community. In the U.S., this role is played primarily by the Institute of Education Science (IES), which offers trial grants and maintains the 'What Works' Clearinghouse.[2] In England, the key system actor is the EEF, an independent foundation jointly funded by government and a major charity, the Sutton Trust.[3] EEF maintains a 'Teaching and Learning Toolkit' which communicates aggregated evaluation results from their own and other trials.
(2) The role of *researchers* is to employ evaluation methods which, within the counterfactual scientific paradigm, can establish the causal impact of a particular programme or set of practices on designated educational outcomes. The primary methodology adopted to do this is the RCT.
(3) The role of *school actors* is to assess the evidence generated by trials and make decisions about allocation of school funds and their own time and energy on the basis of that evidence.

In both the U.S. and the U.K., 'What Works' in education has become associated with the policy drive to reduce 'achievement gaps' between black and white students, or students who are and are not eligible for free school meals. This goal is explicit in the missions of both the U.S. Investing in Innovation fund and the U.K. EEF. This is important context as, if the focus of 'What Works' as a strategy is to support disadvantaged students, it must incorporate methods to move beyond evidence generation to actually improving the conditions of teaching and learning in the schools where those students are. Consequently, there is increased concern about how a 'What Works' strategy can evolve to support the scale of sustainability of promising practices.

This paper starts from the contention that the current approach to 'What Works' inherently limits the potential for impact at scale. There is no shortage of critiques of 'What Works': for its reliance on RCTs (Howe 2004; Morrison 2001; Smeyers and Depaepe 2007; Walters, Lareau, and Ranis 2008) and for representing a hyper-rationalization of educational research, or limiting democratic debate (Oancea and Pring 2008). All of these should be taken seriously, but this proposal starts from the position that randomized evaluations are going to remain a major part of education research, and that a new strategy must incorporate this methodology. This is not without cause; randomized evaluations are less open to corruption and 'pseudo evaluations' (Fowler 2012) and show no evidence of publication bias compared to other impact evaluations (Vivalt, forthcoming). This proposal focuses therefore not on the problems related to randomization but to the particular approach to randomized evaluations that underpins 'What Works', known as 'black box RCTs' (Scriven 1994). This label refers to the fact that trials are not intended to provide information about how, why or under what conditions the programme works – hence, they leave the 'black box' unopened. Although few evaluators still use this label, 'What Works' guidance is in line with this description: the priority in trial design is to give an accurate estimate of the likely impact at scale of an intervention or programmatized approach, not to confirm or refute applied theory (Cook 2002; Torgerson and Torgerson 2013).

Limitations of black box RCTs

Rejection of the black box RCTs in favour of a more theory-based approach to evaluation has a long history in general evaluation literature (e.g. Chen 1990; Weiss 1995). I therefore cover this only briefly to highlight the particular limitations of black box RCTs in the context of education, and in relation to the goals of EEF and I3 grants. To summarize the argument:

- A 'black box' approach to RCTs directs research funding towards more programmatic and less teacher-led 'treatments', and away from many potential forms of improvement.
- Where teacher-led treatments are trialled, the black box model creates uncertainty as to whether an intervention has been implemented as intended, and prohibits learning about how and why an intervention is effective.
- Limited understanding of the how and why of an approach impede proper implementation by others 'at scale'.
- Few interventions that have demonstrated impact in trials are effective at scale, and overall there is little 'take-up' of the results generated by trials.

I now briefly detail each of these points.

Poor fit with teacher-led interventions

Raising student achievement relies on the effective activity of teachers and principals (Hattie and Anderman 2013). Black box RCTs offer very little evidence or information to inform the day-to-day activities of these practitioners, unless they focus on sets of practices as opposed to programmes. The black box approach generates no information about *why* an intervention is successful and therefore how it should best be implemented or enhanced (Bonell et al. 2012). This is the sort of information that practitioners need to make practices 'work' in their setting (Schaffer and Stringfield 2011).

Within the black box approach the teacher is a conduit for the successful delivery of a programme design to the student subjects; very few RCT design take into account the experience of implementers. This gap can be traced to the origins of the 'What Works' strategy in the field of Medicine (Haynes et al. 2012; C4E 2003). Randomized experiments in the medical field have the specific purpose of testing and accrediting specific drugs or (literal) treatments as effective or ineffective. The 'What Works' Clearinghouse and the EEF Toolkit explicitly adopted the same orientation to accreditation, assuming a model of educational change whereby validated educational approaches can be selected and provided to students like products. While this paradigm might suit a small range of educational intervention types, such as computer software or boxed social and emotional learning curricula, it applies only to a narrow portion of potentially helpful educational practices (Howe 2004, 45). Cook's (2002) analysis of the feasibility of RCTs in Education leads him to recommend a focus on interventions where

> treatments are shorter; they require little or no teacher training; … the units receiving different treatments cannot communicate with each other; and when students are the unit of assignment rather than classrooms or whole schools.

Cook recommends interventions such as 'different curricula' or 'new technologies' delivered directly to students. These kind of interventions, however, have been shown to have little or no effect when unsupported by teachers (Higgins, Xiao, and Katsipataki 2012; Kremer, Brannen, and Glennerster 2013).

The alternative to teacher-led interventions is whole-school approaches, often supported by external partners. Trials of such approaches must be treated with care, however, because one cannot assume that there would be sufficient whole schools or external partners who could implement the model at scale in the same way as the initial population (cf. Morgan and Winship 2011, 50).

Weak implementation and scale

'What Works' involves trials of different scales. Most interventions initially undergo an 'efficacy' trial to test if they work under 'ideal' conditions, and then are scaled up in an 'effectiveness trial'. Only one of the EEF effectiveness trials has had definitive impact. Likewise, over 90% of effectiveness trials of educational interventions in the U.S. have resulted in no sign of impact (C4E 2013). This low success rate, around half as many successful outcomes as in phase II clinical trials in medicine (Kane 2015), indicates that a key source of null effects is incomplete or inaccurate implementation when an intervention is moved beyond its initial context. In a response, 'What Works' system actors in both the U.K. and the U.S. are working on improved protocols for reporting on implementation (Dhillon, Darrow, and Meyers 2015; Lendrum and Humphrey 2012). While this is an important and necessary improvement to trial methodology, improved implementation and process evaluation (IPE) considerably raises the cost of each trial. Moreover, developing IPE protocols can be very challenging as there is often limited knowledge of the key ingredients of an intervention, or the necessary conditions for implementation (Darrow and Goodson, forthcoming).

Low adoption

The relatively low number of definitive results, and the restricted range of the educational practices which can be successfully submitted to trial, together limit the evidence that can be generated by randomized evaluations that is useful for schools and teachers. Amongst system leaders, the concern in the 'What Works' strategy is that schools are not 'using' the aggregated knowledge.

Low 'take-up' of research findings among practitioners is a well-known problem in Education (Levin 2013; Walter, Nutley, and Davies 2005). It has been met by generations of efforts to 'bridge the gap' between researchers and practitioners (CUREE 2005; Davies, Nutley, and Walter 2008). Currently, theEEF is funding several trials of different strategies to improve evidence use. These include methods to 'package' findings in more useable ways, or to employ 'research champions' in schools, who can help teachers to identify and understand existing studies (Brown 2015). These are likely to be beneficial projects, but this range of responses is limited because it assumes no possibility of change in the trial or evaluation process itself. Making studies more accessible to teachers does not solve the problem of 'take-up' if the evidence is not clear or strong enough to provide guidance in response to their questions. A complementary effort would be a new approach to evaluation design which would supply output that is more compatible with ongoing teacher-led improvement practices.

A proposal: 'What Works' 2.0

In the above sections, three central weaknesses have been identified:

- Methods developed to evaluate products do not serve well the needs of teachers and school leaders, who need to understand the principles of an approach in order to implement it effectively in their setting.
- Methods used to define products and create implementation protocols are less good at identifying the core elements of an intervention which could be scaled as effective practice.
- Attempts to increase the take-up of evidence only consider how research can be repackaged, not how research design can be reformed to better fit with educational improvement practices.

Evaluators have been worrying about these problems for some time. Varieties of 'theory-based' or 'causal-chain' evaluation have been recommended to overcome some of these weaknesses (Petrosino 2000). Practitioners of these methods have struggled, however, with how to define theory, and with disagreements as to the purpose of defining theory (Astbury and Leeuw 2010; Birckmayer and Weiss 2000). The weakness of theory-based evaluation as an alternative to black box is that, potentially because it has primarily been developed separate from a specific domain such as education, it is still too closely linked to the definition of a particular programme to be implemented (hence to the definition of 'programme theory'), as opposed to really centring on principles which can be adopted and adapted as part of teacher practice. As an alternative, the following sections outline two adaptations to the 'What Works' strategy to address the above weaknesses, based on adapting to education practices which have been developed in other fields.

Component 1: mechanisms experiments

Where black box RCTs evolved from a medical model of randomized trials, in international development, a different tradition of randomized experiments has been developed by economists (Duflo, Glennerster, and Kremer 2006). In an influential guidance paper on the use of randomization in development economics, Duflo and colleagues explicitly highlight 'the necessary use of theory as a guide when designing evaluations and interpreting results' (4). Economists see randomized field experiments as a means to empirically test interventions based on specific principles – principles derived from economic theory or behavioural science (even if not all experiments live up that goal) (Card, DellaVigna, and Malmendier 2011).

In this approach, where possible, randomized experiments are designed as 'mechanism experiments', which are small-scale field experiments to test a specific behavioural theory (Ludwig, Kling, and Mullainathan 2011). A mechanism, within the counterfactual social science paradigm that forms a basis for 'What Works', *explains how* one thing causes something else (Elster 2007, 7–30). The great barrier to wider adoption of mechanism experiments, however, is that this definition is not well defined, and not universally accepted (Hedstrom and Ylikoski 2010; Morgan and Winship 2011, 338–346). A central debate is whether mechanisms can be equated with intermediary variables or whether, as in the realist tradition, mechanisms are mostly 'hidden' and cannot be directly studied (Astbury and Leeuw 2010, 368; cf. Pawson and Tilley 1997).

Within 'What Works' guidance, this debate is left unresolved: mechanisms are often referred to as part of 'logic models', but it is not made clear whether or not mechanisms should be observable (e.g. Darrow and Goodson, forthcoming, 9). In the causal-chain models favoured in 'What Works', mechanisms may be distinguished from 'inputs' or 'intermediate outcomes', but still play a role in the causal effect. Thus where an input were a professional development programme, an intermediate outcomes might be a change in teacher talk brought on by the programme, and the mechanism for improved learning outcomes would be this talk promoting different responses, perceptions and learning behaviours in students. As a focus of experiments, and therefore remaining within a counterfactual paradigm, a workable definition of key mechanisms in education is that mechanisms are the proximate and most significant factor impacting on a change in learning behaviours or understanding.

Because mechanism experiments are designed to test a specific principle, unlike in the black box RCT there is arguably more chance of being confident that a design has been enacted as intended. Moreover, unlike in theory-based evaluation where a whole causal-chain may be under examination, it is less costly and complicated to monitor just one key mechanism of interest. Likewise, a mechanism experiment reduces the uncertainty created by implementation problems by focusing attention on a key change necessary for an intervention to be considered 'implemented'.

Designing mechanism experiments

This section outlines a practical approach to incorporating mechanism experiments into the design of educational trials. Methodologists recognize that a government or group of schools may want to implement a programme, and not wait for an initial test of the central mechanism. In such cases, it is possible to design programme and policy evaluations in a way that they *also* tests central mechanisms (Imai, Tingley, and Yamamoto 2013; Keele, Tingley, and Yamamoto 2015). Yamamoto and colleagues describe four experimental designs by which to estimate the effect of a mediator variable[4] (the 'indirect effect') from the rest of the treatment effect (the 'direct effect'). These designs rely on researchers being able to some extent to manipulate a mediator variable, which may be challenging in educational contexts. For example, in a trial of a programme to improve teacher feedback, a key mechanism might be that teachers employ detailed as opposed to generic feedback from a training session. A manipulation of that mechanism might be that teachers are randomly assigned to receive different forms of initial training. This process, however, might lead to different kind of behaviours than if teachers were naturally responding to a standard training on feedback.

Encouragement designs. As an alternative to direct manipulation, Imai, Tingley, and Yamamoto 2013 propose an 'encouragement' design, whereby subjects are randomly assigned to be encouraged to 'take on' higher values of the mediator (19). The key assumptions of the encouragement design are similar to those of an instrumental variable design: the encouragement must affect the outcome only through the mediator, and the encouragement has a linear effect on the mediator, that is, there are no 'defiers' or respondents who are disproportionately affected. In the example above, an encouragement design might be that teachers are randomly assigned to receive reminders from their senior leadership team on an initial training. In other words, the goal is to create a design where the mediator group receive additional prompting to behave in the way the treatment intends, not a design where the mediation status is effectively a different treatment assignment. Mixed method RCT designs have demonstrated the capability to capture variation in treatment implementation and response, which could check the validity of a given encouragement design (Spillane et al. 2010, 17)

Incorporating encouragement designs into black box RCTs. Using the encouragement design, it is possible to incorporate a mechanism experiment into a black box RCT. Evaluators can work with programme developers to identify the key hypothesized mechanism by which the programme has an effect, which may be an extension of constructing a detailed logic model (Kellogg 2004; Weiss

1998). Schools agree to be randomly allocated to the treatment, control or treatment + encouragement condition. In the trial, all teachers (or other implementers) in treatment schools receive basic training and support as defined by the programme. In the mechanism trial, a randomly selected group of teacher in the treatment group also receive ongoing 'encouragement', in the form of guidance and prompts on the key elements of the programme. This is an attempt to manipulate the key component of the intervention and separately test the impact of this component.

Using encouragement designs, mechanism experiments could be inserted into trials of different interventions. Testing mechanisms across several trials could differentiate between mechanisms associated with teacher behaviours, and those associated with a specific programme. The test that a particular mechanism is generalizable would rest on whether the encouragement state is related to a significant positive indirect effect across multiple treatment states (cf. Birckmayer and Weiss 2000, 410).

This proposed approach to mechanisms differs from the way mechanisms are currently conceived in 'What Works' guidance. Evaluators are working on methods to specify the predicted mechanisms by which an intervention design leads to improved outcomes (Darrow and Goodson, forthcoming, 9). The goal of this effort, however, is to improve *fidelity* of implementation: once these elements are clearly specified, evaluators design indicators to show whether the most important parts of an intervention have been implemented as intended (Dhillon, Darrow, and Meyers 2015). This emphasis on fidelity may not be suited to education (Lendrum and Humphrey 2012). Moreover, this effort is not intended to support a shift towards mechanism experiments because mechanisms are identified to be monitored rather than *manipulated*.

The advantages of mechanisms as the unit of evidence in improving education

Establishing how to effectively manipulate mechanisms in educational contexts would take work, but the effort has the potential to considerably enhance the applicability of outcomes from trials. Testing more discrete mechanisms of educational change should improve the chances of transferability between contexts and populations (Deaton 2010). Moreover, it would contribute to understanding the 'how' and 'why' questions teachers are most interested in (Bennett 2015). This is a vital step towards generating research output which teachers and school leaders can make good use of. The black box approach to trials of programmes assumes that school-based decision-makers can adopt and implement pedagogical programmes at will. There are, however, multiple layers of decision-makers who jointly determine what happens in schools, from Ministers for Education to teachers. It is relatively unlikely that a programme trialled in one context can be implemented in full in another: some adaptation to the local school and system would always be required. Consequently, evidence of causal effects, such as that generated by RCTs, are only useful in education under certain conditions (Caro 2015, 578). Knowledge generated by experiments is more likely to be useful if it is more granular; rather than deciding which pre-packaged programme to use, teachers need detailed pedagogical and pedagogical content knowledge: details of what actions or strategies that will help their students learn (Villegas-Reimers 2003, 39).

Mechanisms and the learning sciences

The central advantage of designing trials around mechanisms is that it would for greater continuity between randomized evaluations and the 'learning sciences'. This field, which has emerged from the study of cognitive science and development psychology and which focuses on the study of how, when and where humans learn best, presents learning as a highly social and emotional process (Bransford et al. 2000; OECD 2007). Consequently, a learner's motivation and attention, which can be influenced by a wide variety of external and particularly social factors, are vital component of improved learning.

Attention to these intra- and interpersonal factors is particularly relevant to 'What Works' strategies. As outlined above, EEF and I3 are focused on closing achieving gaps. Intervention which might be *particularly* effective with groups who are 'behind' are those which tackle the related emotional and motivational implications of being 'behind' (Deakin Crick and Goldspink 2014; Dweck 2006). Attempts to programmatize theories such as 'growth mindsets' into interventions have not been successful (Rienzo, Rolfe, and Wilkinson 2015). Trials which test a core psychological theory need for allow for considerably adaptation of their application, to take into account the fact that the method for shifting a mindset might be highly variable depending on the social and relational context, and the meaning students have made of it (Tabak 2004, 227). Progress in testing promising theories such as growth mindset outside of lab-based settings could particularly benefit from a mechanism experiment as opposed to black box approach.

Part 2: supporting implementation at scale

The need to focus on mechanisms as opposed to programmes comes to the fore in relation to the growth of teacher-led improvement efforts in education. These efforts have developed in response to the failure of 'What Works' to promote systemic improvement, and represent a more practitioner-driven and continuous form of improvement in education (Reimers and Villegas-Reimers 2014). A reformed 'What Works' strategy needs to integrate rather than compete with these developments. Examples of improvement efforts are described here to illustrate how they create an audience for the output of mechanism experiments.

Networked improvement communities

'Improvement Science' is a set of approaches developed and refined in the field of organizational studies (Langley et al. 2009) and currently championed as a high-leverage approach for educational research (Bryk et al. 2015; Lewis 2015). Key aspects of improvement science involve studying educational practice as a system, and attending to variability across contexts, and learning from this variability. Instances of variation represent possibilities to generate insights about what might be causing more improvement in some contexts than others, and consequently insights into the sources of improvement. Improvement science highlights the need to attend to the situation and perspectives of the human actors in involved in bringing about the change (Langley et al. 2009).

Improvement science generates evidence about the interactions and processes which bring about change (improvement), as opposed to evidence about programmes. Improvement Science therefore has similarities with Design-based Implementation Research (DBIR), which studies the 'conditions' under which an intervention works, and has been used effectively in the U.S. to overcome weaknesses in prior attempts to scale interventions (Fishman et al. 2013). Like Improvement Science, DBIR assumes that a high degree of adaptation is required for any whole-scale programme to 'work' in a given context. Researchers using DBIR are now developing approaches to build knowledge of the conditions for successful implementation across different projects (Means and Harris 2013). This approach illustrates the need to learn about the effectiveness of implementation as an ongoing process, as much as of individual programmes (Fowler 2012, 270)

In recent years, increased investment in improvement science in education in the U.S. has led to the establishment of 'Networked Improvement Communities' (Bryk et al. 2015). In these communities, groups of schools (or other organizations, such as community colleges) work together on a commonly agreed problem, and share their learning about the small changes they are making. Gradually, certain changes are accepted by all partners as an 'improvement', while others are rejected as having been tried and failed. Over time, the community develops improved practice, which produces better outcomes for students.

Inquiry networks

Inquiry networks have some similarity with networked improvement communities, but operate in a more diffuse way: not all teachers are focused on the same problem. They also require less input from research partners. This may suit some local authorities better, as they are less expensive to run and are suited to communities where there is a greater diversity of challenges to address. In this model, teachers draw on external evidence to inform processes of local evidence-generation, with the goal of seeking to establish 'What Works' *in their own context*.

An example of this model that has been implemented in multiple contexts is known as Spirals of Inquiry (Timperley, Halbert, and Kaser 2014). This approach involves teachers engaging in cycles of scanning to understand how students are experiencing their education; developing a focus for their inquiry and a 'hunch' about which of *their* actions could change the patterns they are seeing; looking at published evidence to inform their practice; taking action; and finally checking to see if the action is having the desired effect (Halbert and Kaser 2014). Checking might involve surveying students, but the central process involves teachers or administrators holding conversations with students, either one-on-one or in small groups, using questions that have been developed by prior users of the spiral.

This approach has been highlighted by the OECD, in particular for its capability to support the implementation of new learning and teaching approaches at scale (OECD 2015). Moreover, empirical research has evidenced the theories of action underlying spirals of inquiry (McGregor 2013; Timperley, Parr, and Meisel 2010).

Incorporating improvement and inquiry networks into 'What Works'

The growth of network approaches offers an important opportunity to 'What Works': Network approaches complement a 'What Works' strategy where new evidence about factors which improve learning is incorporated into ongoing cycles of teacher self-development. As illustrated by the example of the Networks of Innovation and Inquiry, when teachers are involved in ongoing inquiry work, they are more likely to find out about new evidence and to incorporate into their practice in an ongoing way.

Network approaches not only increase the likelihood of evidence use, but increase the quality of implementation. By building on traditions of formative evaluation, improvement science and inquiry networks build in feedback loops, which greatly increase the likelihood implementation will be effective (Argyris and Schon 1974; Fowler 2012). Feedback loops – in the form of teachers studying how their students are responding to a new practice or intervention – focus attention on the real mechanisms of change: the change in the learner's perception of their environment and consequently in their behaviour. The learner becomes the focus as opposed to the protocols of the programme to be implemented. When teachers are part of large networks engaging in this process, these new insights could even feedback back into ideas about new core mechanisms to be tested experimentally.

RCTs and networks are suitable for different purposes or stages in a research process (cf. Nelson and O'Beirne 2014; Nutley, Walter, and Davies 2009). Proponents of the above approaches recognize the need for 'efficacy research', which focuses on establishing causal claims with high internal validity, and supplied the first stage of ideas that can be developed at scale (Power et al. 2005; Sabelli and Dede 2013). Currently, they are reliant on collections which do not define actions in a granular way (e.g. Hattie 2008). Randomized evaluations of transferable mechanisms can effectively inform network activities because they are designed to generate causal estimates that are robust and reliable. The two adaptations, mechanism experiments and improvement networks, as opposed to being in opposition are highly compatible.

Starting points for defining mechanisms

The full realization of a shift towards 'What Works 2.0' would involve redefining or complementing the content of collections such as the U.S. What Works Clearinghouse and the EEF Teaching and Learning Toolkit in terms of mechanisms. As described above, the main barrier to this effort is likely to be the historical difficulty with defining mechanisms. A second potential barrier is that approaches developed in international development (mechanism experiments) and public health (improvement science) will prove impossible to adapt to education, due to the lower availability of research funds in the education field. The final section of this proposal therefore sets out several initial steps for developing expertise in defining mechanisms within the education field, including identifying some possibilities for reallocating existing funding streams.

1. Learn from other fields

Other fields, including public health, have developed expertise on identifying mechanisms. Advances in approaches to clarifying causal mechanisms are proceeding in the medical field, following new guidelines in the U.K. for evaluating complex medical interventions (Craig et al. 2008; Moore et al. 2015). In particular, practitioners of realist evaluation (Pawson and Tilley 1997), who work predominantly in social policy and public health, have developed expertise on tracing mechanisms. Realist evaluations are designed to test a set of hypotheses each in the form of a 'context, mechanism, outcome configuration' (Pawson 2013, 21). CMOs are multiple, because there any programme or approach includes multiple components and stimuli, 'creating different resources which trigger different reactions among participants' (22). Realist evaluators are trained to seek out mechanisms that are 'unspoken', and small 'nudge'-like features that shape decision-making and behaviour (Pawson 2013, 127–128). While there are significant ontological assumptions underpinning realist evaluation that cannot be overlooked, elements of the expertise should transfer to non-realist or pragmatist practitioners.

An initial part of the 'What Works 2.0' strategy could be to pool events funding from across the U.K. What Works Centres to fund a small number of researcher inquiry groups, bringing together practitioners from public health, social policy and education to share expertise on identification of mechanisms for testing.

2. Extract learning from partnership approaches

As outlined above, in implementation and inquiry networks, practitioners and researchers engage in adapting a programme or practice to get it to 'work' in a context. This activity entails making implicit 'conjectures' about what is causing improvement for students – in other words, of the key mechanisms of educational change in that context (Sandoval 2004). Likewise, careful post-hoc analysis of collections of trials can help to identify core components across programmes (Glennerster 2015; Jones et al. 2015). 'What Works' system actors could take on a new role as co-ordinating bodies to systematically capture and refine these conjectures across networks and bodies of existing trials. This would involve re-writing some project manager job descriptions to include a 'learning' and collation capacity.

3. Create established methods for reporting mechanisms

Applied researchers in several fields have identified a problem in establishing replicable methods for reporting mechanisms, or 'core ingredients' (Maggin and Johnson 2015; Means and Harris 2013, 265; Michie and Johnston 2012). In the field of behaviour change, focused on health interventions, an international effort has led to the development of a taxonomy of researched Behaviour change techniques, developed to meet the need for a 'consistent nomenclature' for describing 'active ingredients' (Michie et al. 2013). The taxonomy has already received over 250 citations.

What Works centres such as EEF would be well-placed to take on the role of developing a similar taxonomy of key mechanisms of educational change: techniques, behaviours, environmental alternations and tools which improve learning. Learning from methods for categorizing interventions in the field of school psychology, it would be important to categorize mechanisms at different levels of generality, to distinguish where there is evidence for efficacy at a general or population or context-specific level (Quintana and Atkinson 2002; Wampold, Lichtenberg, and Waehler 2002). The project team working on further development of the toolkit could be re-directed to plan a project for the next iteration of the toolkit to be composed in terms of 'active ingredients'.

Conclusion

This paper outlines a proposal for two adaptations to the strategy known as 'What Works', aimed at increasing the potential of the approach to fulfil its goal of improving the educational outcomes of disadvantaged students. The new strategy entails designing RCTs to incorporate mechanism experiments; rebuilding aggregation hubs around a new 'unit' of evidence in the form of educational change mechanisms; and surrounding these hubs with an infrastructure of improvement and inquiry networks. The shared set of mechanisms would provide a means to learn from, and form a continuity with, the insights generated within improvement and inquiry networks, and within the learning sciences. The work of developing mechanism 'labels' would orient 'What Works' efforts away from programmatic interventions that are rarely implemented, and towards a higher leverage and more reliable type of 'what': the factors that are closest to an impact on learning. The mechanism labels would embed within 'What Works' efforts the insight that closing educational attainment gaps necessarily involves social and psychological factors such as mindset shifts and shifts in the perceptions of others.

This proposal outlines the first steps for implementing this strategy by using the convening power of the What Works centres to facilitate learning across and between public policy fields; by acting as a central organizer for diverse network initiatives; and by holding the responsibility for developing an educational change taxonomy.

Notes

1. Internationally in the education field, 'What Works' is predominantly associated with RCTs; in the U.K., the 'What Works Network', recently established by the UK government and funded by the Economic and Social Research Council, uses the term more broadly refers to the sharing and use of 'high quality evidence for decision making'. See www.gov.uk/guidance/what-works-network.
2. http://ies.ed.gov/ncee/wwc/.
3. https://educationendowmentfoundation.org.uk/about/.
4. Not all mediator variables are mechanisms (Vancouver and Carlson 2015), but these designs are intended to isolate mediators that act as mechanisms, as in, have a considerable causal impact on the intended outcome.

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